

		$\sin(20000000) = 0(20000000)$; $\sin(10) = 210(10)(0.174)$ $F = 12.87 = 13 \text{ N (to the right)}$
8	B	Total output power = power in doing work against friction + power in gaining gravitational potential energy $P = Fv \rightarrow \text{resistive force} = \frac{P}{v} = \frac{1.0}{0.10} = 10 \text{ N}$ Power in overcoming friction = $Fv = 10 \times 0.20 = 2 \text{ W}$ Power in gaining gravitational potential energy = $mg \times \text{vertical component of speed} = 0.500 \times 9.81 \times 0.20 \times \sin 30^\circ = 0.49 \text{ W}$ Total output power = 2.5 W
9	D	Power from generator = $VI = 22 \times 220 = 7260 \text{ W}$